EMBEDDED INDOOR AND OUTDOOR AIR QUALITY MONITORING VENTILATION SYSTEM

MINI PROJECT REPORT

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CERTIFICATE

This is to certify that the report entitled **"EMBEDDED INDOOR AND OUTDOOR AIR QUALITY MONITORING AND VENTILATION"** is a bonafide report of the mini project (CSD 334) presented during VIth semester by **ABIN VARGHESE, ALAN PAUL JOHN, AMAL SUNNY, ARYADATH L S** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Computer Science and Engineering.

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ABSTRACT

Poor indoor air quality has become a significant concern due to enclosed indoor environments, limited air circulation, and prolonged human occupancy. Accumulation of carbon dioxide (CO_2), smoke, and other harmful gases inside rooms can lead to health issues such as headaches, fatigue, and respiratory discomfort. Therefore, there is a need for a reliable and cost-effective system to monitor indoor air quality and take timely corrective action.

This project presents an **Embedded Indoor Air Quality Monitoring and Intelligent Ventilation System** designed to monitor and improve indoor air conditions using low-cost sensors and automated control logic. The system employs air quality sensors to measure both indoor and outdoor air quality levels, providing an indirect indication of CO_2 concentration and harmful gaseous pollutants. The sensed data is processed using a microcontroller programmed in C / Python, enabling real-time comparison of indoor and outdoor air quality.Based on this comparison, the system intelligently controls a ventilation mechanism. When indoor air quality is poor and outdoor air quality is relatively better, ventilation is enabled to expel polluted indoor air, and the incoming air is passed through an activated carbon filter to reduce gaseous pollutants. In cases where both indoor and outdoor air quality are poor, external ventilation is disabled, and the system switches to internal air circulation with activated carbon filtration while indicating the condition to the user. Sensor readings and system states are displayed locally using a serial monitor or display unit, eliminating the need for web-based monitoring. The proposed system is cost-effective, simple, and suitable for academic implementation, making it applicable for homes, classrooms, offices, and other indoor spaces.

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CHAPTER 1
INTRODUCTION

1.1 Overview

The development of embedded indoor and outdoor air quality monitoring and ventilation control systems requires the integration of sensing, processing, and automated control mechanisms. This project focuses on designing an Embedded Indoor and Outdoor Air Quality Monitoring and Ventilation Control System using the NodeMCU microcontroller. The system uses MQ-135 gas sensors placed indoors and outdoors to detect air quality conditions. The NodeMCU processes the digital signals from the sensors and controls relay modules to operate ventilation fans accordingly. Based on the air quality status, the system automatically activates appropriate ventilation to improve indoor air circulation without manual intervention.

1.2 Problem Statement

Poor indoor air quality has become a significant concern due to the accumulation of harmful gases, smoke, and inadequate ventilation in enclosed environments such as homes, classrooms, and offices. The primary challenge is to design a system capable of continuously monitoring indoor and outdoor air conditions and responding appropriately to maintain a healthy indoor environment. The problem is to develop an embedded system that can detect variations in air quality using gas sensors, process real-time sensor data, and automatically control ventilation mechanisms to regulate airflow. The system must operate reliably using sensor feedback and simple control logic to improve indoor air circulation without requiring manual intervention.

1.3 Objectives

- **Hardware Integration:** To interface a NodeMCU (ESP8266) microcontroller with MQ-135 gas sensors, relay modules, ventilation fans, and a regulated power supply system for reliable air quality monitoring and control.
- **Air Quality Detection:** To monitor indoor and outdoor air conditions using MQ-135 sensors that provide digital signals indicating whether the air quality is safe or polluted.
- **Control Logic Implementation:** To develop a decision-based control mechanism that processes sensor signals and determines the appropriate ventilation response.

- **Automated Ventilation Control:** To operate exhaust and intake fans through relay modules to regulate air circulation based on the detected air quality conditions.
- **Indoor Air Quality Improvement:** To automatically improve indoor air conditions by removing polluted air and allowing filtered air circulation without requiring manual intervention.

1.4 Scope and Limitations

The scope of this project includes the design and implementation of an embedded system capable of monitoring indoor and outdoor air quality and automatically controlling ventilation based on sensor feedback. The system integrates MQ-135 gas sensors, a NodeMCU microcontroller, relay modules, and ventilation fans to detect air quality conditions and regulate airflow accordingly. The entire system operates using a dedicated power supply with voltage regulation provided through a buck converter.

However, the current implementation has certain engineering limitations:

- **Limited Gas Detection Accuracy:** The MQ-135 sensors provide general indications of air quality levels rather than precise measurements of specific gases such as CO_2 or CO. The system therefore identifies air quality trends rather than exact pollutant concentrations.
- **Binary Air Quality Output:** The sensors are used in digital mode, which provides only two states (safe or polluted). As a result, the system cannot measure gradual changes in air quality levels.
- **No Advanced Air Purification:** The system primarily improves air conditions through ventilation and basic filtration. It does not include advanced purification technologies such as HEPA filtration or chemical air treatment.
- **No Remote Monitoring:** The current system operates locally and does not include remote monitoring or cloud-based data visualization capabilities.

CHAPTER 2
LITERATURE REVIEW

2.1 Background Study

Indoor air quality monitoring has become an important area of research in environmental sensing and smart building technologies. Increasing urbanization and the widespread use of enclosed spaces have led to growing concerns about the accumulation of harmful gases and poor ventilation in indoor environments. Sensor-based monitoring systems have therefore become essential for detecting air quality conditions and maintaining healthy living spaces. However, in low-cost embedded systems where precise environmental sensing equipment is not available, simple gas sensors combined with automated ventilation mechanisms provide an effective and practical solution for improving indoor air conditions through continuous monitoring and controlled air circulation.

2.2 Microcontroller Comparison

Traditionally, microcontrollers such as the Arduino UNO (ATmega328P) have been widely used for environmental monitoring and basic embedded applications. However, the NodeMCU ESP8266 was selected for this project due to its integrated Wi-Fi capability, higher processing speed, and greater memory resources compared to conventional microcontrollers. The ESP8266 operates at a clock speed of up to 80 MHz and includes built-in networking support, making it suitable for smart environmental monitoring systems. Additionally, the NodeMCU provides sufficient GPIO pins for interfacing sensors and relay modules while maintaining efficient real-time processing of sensor signals. These features make it a practical and cost-effective choice for implementing an embedded air quality monitoring and automated ventilation control system.

2.3 Air Quality Monitoring Approaches Comparison

Various methods exist for monitoring and improving indoor air quality, each with different hardware complexity and cost requirements.

- **High-Precision Environmental Monitoring Systems:** These systems use advanced sensors such as NDIR CO_2 sensors, particulate matter sensors, and professional air quality analyzers. While they provide highly accurate measurements, they require expensive hardware and complex calibration, making them less suitable for low-cost embedded systems.

- **IoT-Based Cloud Monitoring Systems:** Many modern air quality systems use multiple sensors connected to cloud platforms for real-time monitoring and data analytics. Although these systems provide remote monitoring and historical data analysis, they require continuous internet connectivity and additional infrastructure.

- **Embedded Sensor-Based Monitoring Systems:** These systems utilize simple gas sensors combined with microcontrollers to detect air quality conditions and perform automated responses such as ventilation control. They require minimal computational resources and are cost-effective while still providing reliable air quality trend detection. Due to their simplicity and low hardware requirements, this approach is well suited for embedded air quality monitoring and automated ventilation control systems.

2.4 Review of Related Works

Indoor air quality (IAQ) has emerged as a critical area of research due to its significant impact on human health, productivity, and overall well-being. Modern lifestyles result in individuals spending a majority of their time indoors, where air pollution levels can often exceed those found outdoors. Common indoor pollutants include particulate matter (PM), volatile organic compounds (VOCs), carbon monoxide (CO), carbon dioxide (CO_2), and other toxic gases. Prolonged exposure to these pollutants can lead to respiratory disorders, headaches, fatigue, and long-term health complications. Hence, continuous monitoring and effective control of indoor air quality have become essential.

Several advanced approaches have been proposed for monitoring and managing IAQ. Many systems employ multiple sensors to measure environmental parameters such as gas concentration, temperature, and humidity. These systems are often integrated with Internet of Things (IoT) technologies, enabling real-time data acquisition, wireless communication, and cloud-based monitoring. Users can access air quality data remotely through mobile or web applications, allowing for better awareness and control. However, such systems typically require complex architectures, continuous internet connectivity, and higher implementation and maintenance costs, making them less suitable for low-cost and small-scale applications.

In contrast, low-cost gas sensing technologies have gained significant attention in recent years. Metal oxide semiconductor (MOS) sensors, such as the MQ series, are widely used due to their high sensitivity, compact size, ease of interfacing, and affordability. These sensors are capable of detecting a wide range of harmful gases and are commonly integrated with microcontrollers in embedded systems. Their simplicity and cost-effectiveness make them ideal for real-time air quality monitoring in residential and small-scale environments.

Ventilation is one of the most effective methods for improving indoor air quality. Proper ventilation helps in removing contaminated indoor air and replacing it with

fresh outdoor air, thereby reducing pollutant concentration. Mechanical ventilation systems, particularly fan-based systems, are widely used for controlled airflow management. These systems can be automated based on sensor inputs, ensuring efficient and timely response to changes in air quality. Effective ventilation not only enhances air quality but also improves thermal comfort and energy efficiency.

Various air purification techniques, including filtration, adsorption, and chemical treatment, have also been developed to improve indoor air conditions. While these methods provide high efficiency in pollutant removal, they often increase system complexity, cost, and maintenance requirements. Therefore, for cost-sensitive and compact systems, ventilation-based approaches offer a simpler and more practical solution.

Based on the analysis of existing methods, it is evident that there is a need for a cost-effective, reliable, and easy-to-implement solution for indoor air quality monitoring and control. Hence, this project proposes an embedded system that utilizes low-cost MQ-135 gas sensors and a NodeMCU microcontroller for real-time monitoring. The system automatically controls ventilation using relay-driven fans, ensuring improved indoor air conditions without the need for complex infrastructure. This approach provides a balanced solution in terms of performance, cost, and simplicity.

CHAPTER 3
SYSTEM DESIGN

3.1 Hardware Architecture

The hardware architecture of the Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System is designed using a modular approach consisting of sensing, control, actuation, and power subsystems. The system is centered on the NodeMCU (ESP8266) microcontroller, which coordinates sensor inputs and control outputs to regulate ventilation.

MQ-135 gas sensors are used to detect air quality conditions in both indoor and outdoor environments. The sensor signals are processed by the microcontroller to determine the air quality status. Based on this information, relay modules control the operation of ventilation fans to improve air circulation.

A regulated power supply with a buck converter ensures stable operation of all electronic components within the system.

3.1.1 System Overview

The system follows an embedded monitoring and ventilation control architecture using gas sensors and automated airflow control mechanisms. Two MQ-135 gas sensors are used for detecting indoor and outdoor air quality conditions, enabling the system to determine whether the air is safe or polluted.

The NodeMCU (ESP8266) microcontroller acts as the central control unit that reads the digital outputs from the sensors and processes the air quality status. Based on these sensor signals, relay modules are used to control ventilation fans that regulate air circulation inside the monitored environment.

The hardware architecture of the system is divided into the following major subsystems:

- **Microcontroller Unit (MCU):** NodeMCU ESP8266 responsible for processing sensor signals and controlling system operations.
- **Sensor System:** MQ-135 gas sensors used to detect indoor and outdoor air quality conditions.
- **Actuation System:** Relay modules and ventilation fans used to control airflow and regulate indoor air circulation.
- **Power System:** A 12V power supply with a buck converter used to provide stable voltage for the microcontroller, sensors, and relay modules.

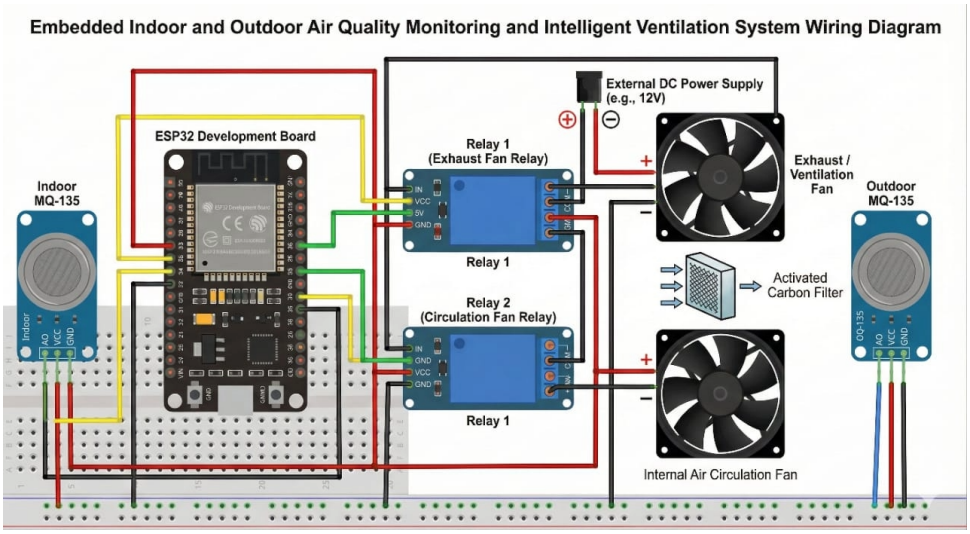


Figure 3.1: Wiring Diagram of the Embedded Indoor and Outdoor Air Quality Monitoring System

3.1.2 Microcontroller Unit

The system uses the NodeMCU (ESP8266) as the primary control unit. It is responsible for:

- Reading the digital signals from the MQ-135 gas sensors placed in indoor and outdoor environments.
- Processing the air quality status based on the sensor outputs.
- Controlling relay modules that operate the ventilation fans.
- Managing the overall decision-making logic for automated air circulation.

The NodeMCU was selected for this project due to its compact design, sufficient GPIO availability, built-in Wi-Fi capability, and compatibility with the Arduino IDE development environment. Its processing capability allows efficient real-time monitoring of sensor data and reliable control of the ventilation system.

3.1.3 Actuation System

The actuation system consists of two ventilation fans used to regulate air circulation within the monitored environment. These fans are controlled using relay modules that act as electronic switches between the microcontroller and the power supply.

The relay modules receive control signals from the NodeMCU microcontroller, which determines when to activate or deactivate the fans based on the air quality signals obtained from the sensors. The relay module receives:

- Control signals from the NodeMCU GPIO pins
- Power supply connections for switching the external load
- Output connections that control the operation of the ventilation fans

The fans are powered using a 12V DC power supply, while the relay modules allow safe switching of the higher voltage load using low-voltage signals from the microcontroller. This arrangement enables automatic activation of ventilation when poor air quality is detected.

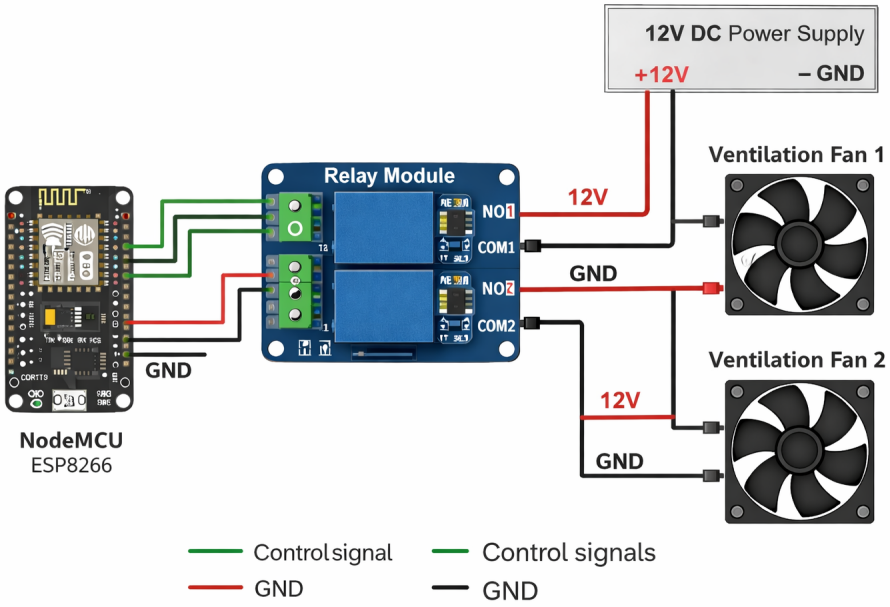


Figure 3.2: Relay Module and Ventilation Fan Connection Diagram

3.1.4 Sensor System

The system uses two MQ-135 gas sensors to monitor air quality conditions in both indoor and outdoor environments. The sensors are positioned as follows:

- **Indoor Sensor** – Detects air quality inside the room.
- **Outdoor Sensor** – Detects air quality outside the monitored area.

Each MQ-135 sensor provides a digital signal indicating whether the detected air quality is within a safe range or polluted. These signals are read by the NodeMCU microcontroller, which processes the sensor outputs and determines the appropriate ventilation response.

Based on the sensor readings, the microcontroller controls the relay modules to activate ventilation fans that regulate air circulation. This enables the system to automatically respond to changes in air quality and maintain a healthier indoor environment.

Although the MQ-135 sensor can also provide analog outputs for measuring gradual changes in air quality, the current system utilizes the digital output mode for simplified detection of safe or polluted air conditions.

3.1.5 Mechanical Layout

The system is implemented within a demonstration house model that represents an enclosed indoor environment. The components are arranged to clearly separate the indoor and outdoor sensing areas while allowing controlled air circulation through the ventilation system. The MQ-135 gas sensors are positioned such that one sensor monitors indoor air conditions and the other monitors outdoor air quality.

Ventilation fans are installed on the walls of the model to regulate airflow between the indoor and outdoor spaces. One fan is used for exhausting polluted indoor air, while the other allows fresh air to circulate into the room.

The electronic components, including the NodeMCU microcontroller, relay modules, and power circuitry, are placed outside the model structure to ensure accessibility and safe wiring. This arrangement simplifies hardware maintenance and prevents damage to the internal airflow path.

This mechanical layout enables effective monitoring of air quality conditions and demonstrates the automatic ventilation mechanism used to improve indoor air circulation. The demonstration model clearly illustrates the interaction between sensing, processing, and ventilation control in the embedded air quality monitoring system.

3.1.6 Power Architecture

The power architecture of the Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System is designed to ensure stable voltage regulation and reliable operation of all electronic components. The system operates using a 12V DC power supply which provides sufficient power for the ventilation fans and the control circuitry.

The power architecture uses a buck converter to step down the 12V supply to 5V, which is required for the NodeMCU microcontroller, MQ-135 sensors, and relay modules. This voltage regulation ensures that all components receive appropriate operating voltage and prevents damage due to overvoltage.

The architecture separates the higher power fan supply from the regulated low-voltage control circuitry while maintaining a common ground reference across the entire system. This arrangement improves system stability and ensures proper operation of both the sensing and actuation subsystems.

Power Supply System

The primary power source for the system is a 12V DC power supply. This supply provides sufficient power for operating the ventilation fans and other electronic components used in the system.

The power supply serves as the main energy source for both the actuation system and the control circuitry. A common ground reference is maintained across all components to ensure stable operation of the microcontroller, sensors, and relay modules.

Ventilation Fan Power Supply

The ventilation fans operate directly using the 12V DC power supply. These fans are connected through relay modules that act as electronic switches controlled by the NodeMCU microcontroller.

This configuration ensures:

- Adequate power for operating the ventilation fans
- Reliable air circulation for removing polluted air and bringing fresh air
- Safe isolation between high-power fan circuits and low-voltage control circuitry

The fan power ground is connected to the common system ground to maintain a stable electrical reference throughout the system.

Logic Voltage Regulation

The NodeMCU microcontroller and sensors require a lower operating voltage compared to the ventilation fans. Since the system uses a 12V power supply, a buck converter is used to regulate the voltage to 5V for the control circuitry.

The buck converter operates as follows:

- **Input:** 12V from the main power supply
- **Output:** Regulated 5V
- **Supplies:** NodeMCU (via VIN pin), MQ-135 sensors, and relay modules

This regulated power supply ensures stable operation of the microcontroller and prevents overvoltage damage to the low-voltage electronic components.

Power Switching System

A power switch is placed between the 12V power supply and the system load components such as the buck converter and ventilation fans. This switch allows the entire system to be turned ON or OFF safely during operation.

Key purposes of the switching system include:

- Safe startup and shutdown of the system
- Controlled power distribution to electronic components
- Prevention of unintended current flow during maintenance or setup

The power switch isolates the load components while maintaining a stable power source for the system.

Power Distribution

The power distribution of the system is designed to provide appropriate voltage levels to different components based on their operating requirements. The system uses a 12V DC power supply as the primary energy source.

The 12V supply directly powers the ventilation fans through relay modules, which act as switching interfaces controlled by the NodeMCU microcontroller.

A buck converter is used to step down the 12V supply to a regulated 5V output. This 5V supply is used to power the MQ-135 air quality sensors used for indoor and outdoor air monitoring.

The NodeMCU microcontroller is powered through the USB connection during system operation and development. The relay modules receive control signals from the NodeMCU GPIO pins and are powered through the 3.3V supply provided by the NodeMCU.

All components share a common ground reference, ensuring stable communication between sensors, relays, and the microcontroller while preventing floating signals within the system.

Grounding Strategy

All subsystems share a common ground reference to ensure signal stability and reliable communication between components.

Common ground connections include:

- 12V power supply ground
- Buck converter ground
- NodeMCU ground
- Relay module ground
- MQ-135 sensor ground

Maintaining a common ground prevents floating references and ensures proper operation of control signals and sensor readings.

3.1.7 Software Architecture

The software architecture of the Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System follows a modular structure to ensure clarity, maintainability, and reliable operation. The system software is developed using the Arduino IDE with the NodeMCU ESP8266 board support package.

The architecture separates sensor data acquisition, air quality processing, and ventilation control into organized functional layers.

Development Environment

- **Programming Platform:** Arduino IDE
- **Microcontroller Platform:** NodeMCU (ESP8266)
- **Programming Language:** Embedded C / C++

The Arduino IDE provides the necessary tools and libraries for programming the NodeMCU and interfacing with sensors and relay modules.

Software Layer Structure

The software is divided into the following logical layers:

- Hardware Abstraction Layer
- Sensor Processing Layer
- Decision Control Layer
- Actuation Control Layer
- Main Control Loop

This layered approach simplifies debugging and enables independent testing of different system components.

Hardware Abstraction Layer

This layer directly interfaces with the physical hardware components. It includes:

- GPIO pin configuration for sensors and relays
- Initialization of sensor input pins
- Relay output control configuration

This layer provides the basic hardware interface required for sensor reading and relay switching.

Sensor Processing Layer

This layer is responsible for:

- Reading digital outputs from MQ-135 sensors
- Detecting indoor and outdoor air quality conditions
- Interpreting sensor signals as safe air or polluted air

Example logic:

- If sensor output indicates safe air $\rightarrow$ Normal condition
- If sensor output indicates polluted air $\rightarrow$ Ventilation required

The sensor processing layer continuously monitors air quality conditions.

Ventilation Control Layer

The ventilation control layer translates sensor decisions into relay control actions.

Implemented control functions include:

- Turn ON exhaust fan
- Turn ON intake fan
- Turn OFF ventilation system

These operations are executed through relay modules connected to the NodeMCU.

Decision Logic Layer

The decision layer processes indoor and outdoor air quality conditions and determines the appropriate ventilation response.

Decision logic:

- If indoor air is good → Fans OFF
- If indoor air is bad and outdoor air is good → Exhaust and intake fans ON
- If indoor air is bad and outdoor air is bad → Filtered air circulation activated

This logic ensures automated ventilation control based on real-time air quality conditions.

Main Control Loop

The main program loop performs the following sequence:

- Read indoor air quality sensor
- Read outdoor air quality sensor
- Process air quality status
- Apply ventilation decision logic
- Control relay modules to operate fans
- Repeat continuously

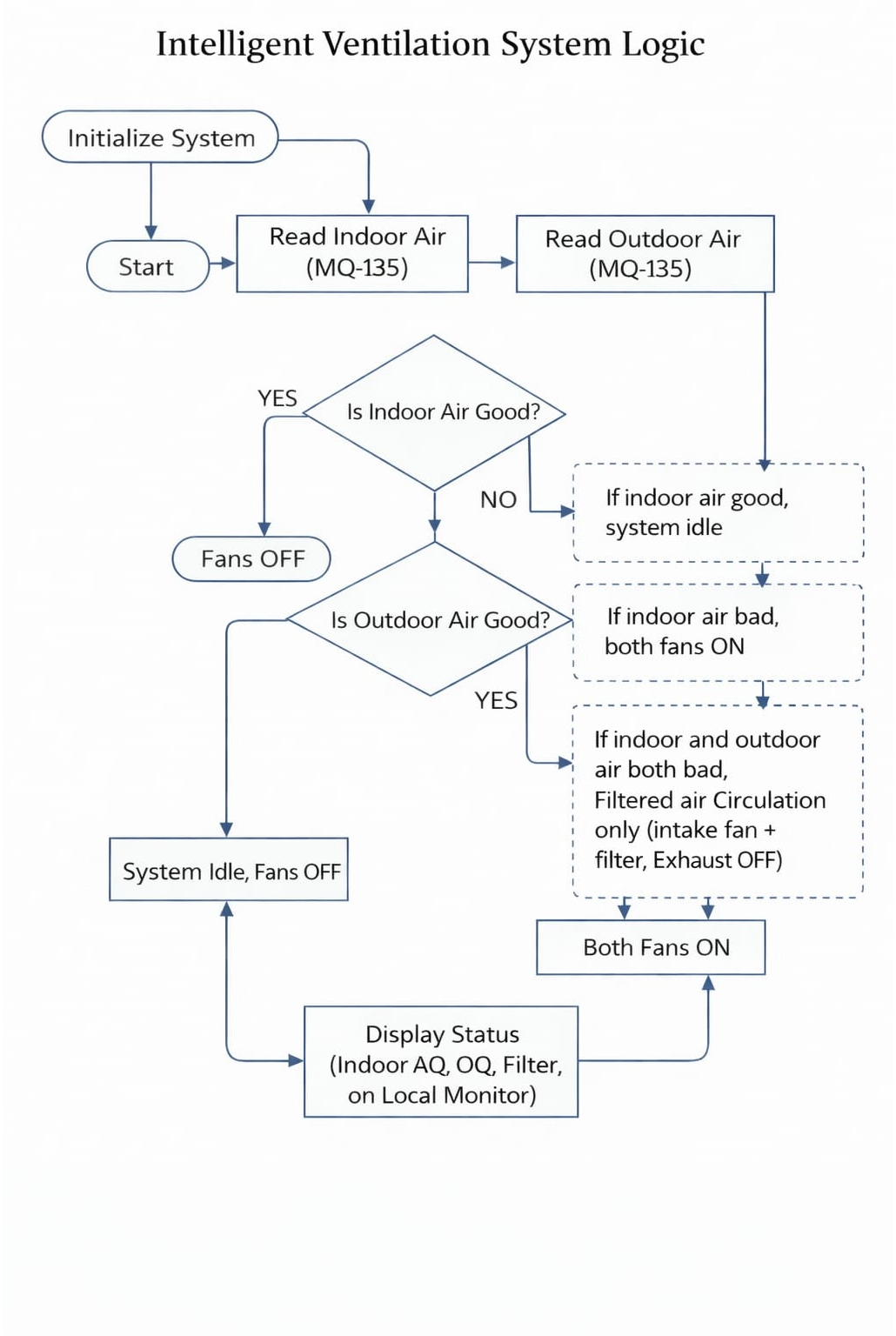


Figure 3.3: Software Execution Flow of the Air Quality Monitoring System

3.1.8 Algorithm Design

The system uses a simple decision-based air quality monitoring algorithm to regulate ventilation. The algorithm continuously monitors indoor and outdoor air quality conditions and automatically controls ventilation fans accordingly. The control logic is implemented within the NodeMCU microcontroller and operates in a continuous loop, allowing the system to respond to changes in air quality in real time.

Overview of Air Quality Control Algorithm

The system operates based on real-time sensor signals obtained from MQ-135 gas sensors placed both indoors and outdoors. These sensors detect the presence of harmful gases and provide digital outputs indicating whether the air quality is safe or polluted.

The NodeMCU reads these sensor signals and processes them using a predefined decision logic to determine the appropriate ventilation response. Based on the detected conditions, the system activates or deactivates ventilation fans to improve indoor air circulation.

This algorithm is particularly suitable for:

- Low-cost embedded monitoring systems
- Real-time air quality detection
- Automated ventilation control

Decision Logic

At every iteration of the control loop, the system performs the following sequence:

1. Read indoor air quality sensor status
2. Read outdoor air quality sensor status
3. Determine overall air quality condition
4. Apply ventilation control rules

The decision rules used by the system are:

- If indoor air is safe → Ventilation fans remain OFF
- If indoor air is polluted and outdoor air is safe → Both intake and exhaust fans are activated
- If indoor air is polluted and outdoor air is polluted → Filtered air circulation mode is activated

This logic ensures that the system automatically improves indoor air quality while avoiding the intake of polluted outdoor air.

Algorithm Characteristics

The proposed air quality monitoring and ventilation control system uses a simple reactive control algorithm that operates based on real-time sensor inputs. The algorithm continuously evaluates the indoor and outdoor air quality conditions and determines the appropriate ventilation response without requiring historical data storage.

Key characteristics of the algorithm include:

- Type: Reactive control algorithm
- Memory Requirement: None
- Air Quality Measurement: Digital sensor output
- Computational Complexity: Low
- Implementation Difficulty: Low

The algorithm operates solely on real-time sensor input and does not store previous air quality data. This makes the system efficient and suitable for implementation on low-cost embedded microcontrollers such as the NodeMCU.

3.1.9 Block Diagrams & Circuit Schematics

This section presents the visual representations of the system architecture and electrical connections used in the Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System. These diagrams illustrate the interaction between different hardware components including the microcontroller, sensors, relay modules, ventilation fans, and power supply units.

The block diagrams help in understanding the flow of signals and control within the system, while the circuit schematics provide detailed information about the electrical connections between components.

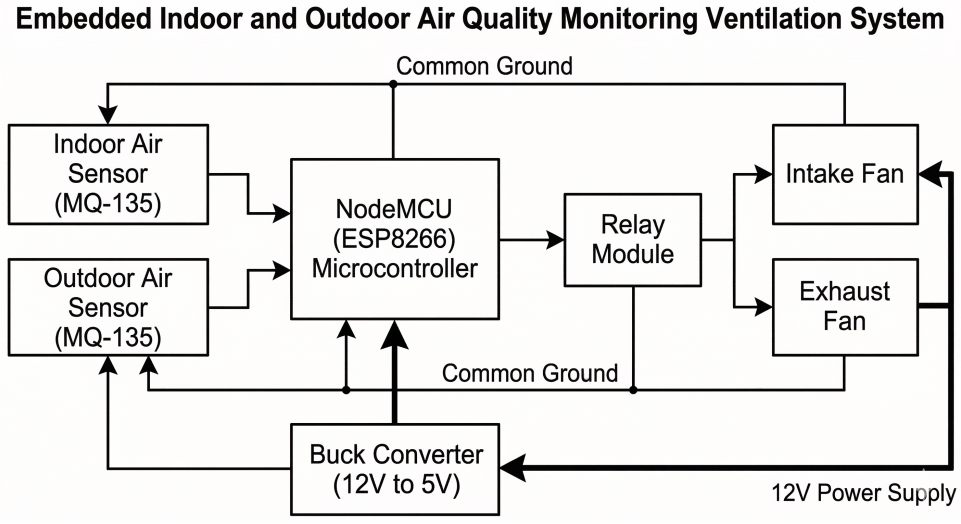


Figure 3.4: Diagram of the Air Quality Monitoring System

Overall System Block Diagram

The block diagram represents the interaction between the major subsystems of the system:

- MQ-135 sensors detect air quality
- NodeMCU processes sensor signals
- Relay modules receive control signals
- Ventilation fans regulate airflow
- Power supply and buck converter provide regulated voltage

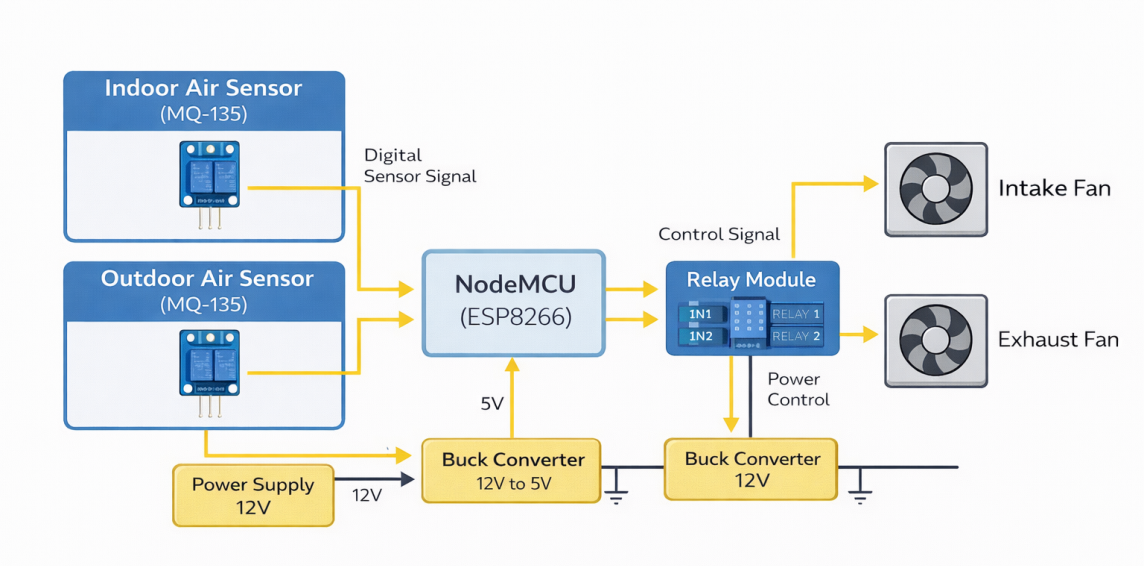


Figure 3.5: Overall System Block Diagram of the Air Quality Monitoring Ventilation System

Working of the System: The system continuously monitors air quality using MQ-135 sensors, which detect gases such as CO_2, NH_3, smoke, and benzene and produce a digital value. This signal is processed by the NodeMCU (ESP8266), where it is compared with predefined threshold levels. If the air quality value remain 1, no action is taken. However, when the air quality value is 0, the NodeMCU generates a control signal to the relay module. The relay acts as a switch to activate the ventilation fans, which help in removing polluted air and circulating fresh air. Once the air quality returns to normal levels, the system automatically turns off the fans. A regulated power supply along with a buck converter ensures that appropriate voltage levels are maintained for all components.

Power Distribution Diagram

This diagram shows the power flow and voltage regulation used in the system.

- 12V power supply
- Buck converter providing regulated 5V output
- NodeMCU powered through VIN pin
- Sensors powered by regulated supply
- Relay modules controlling ventilation fans
- Common ground connection shared by all components

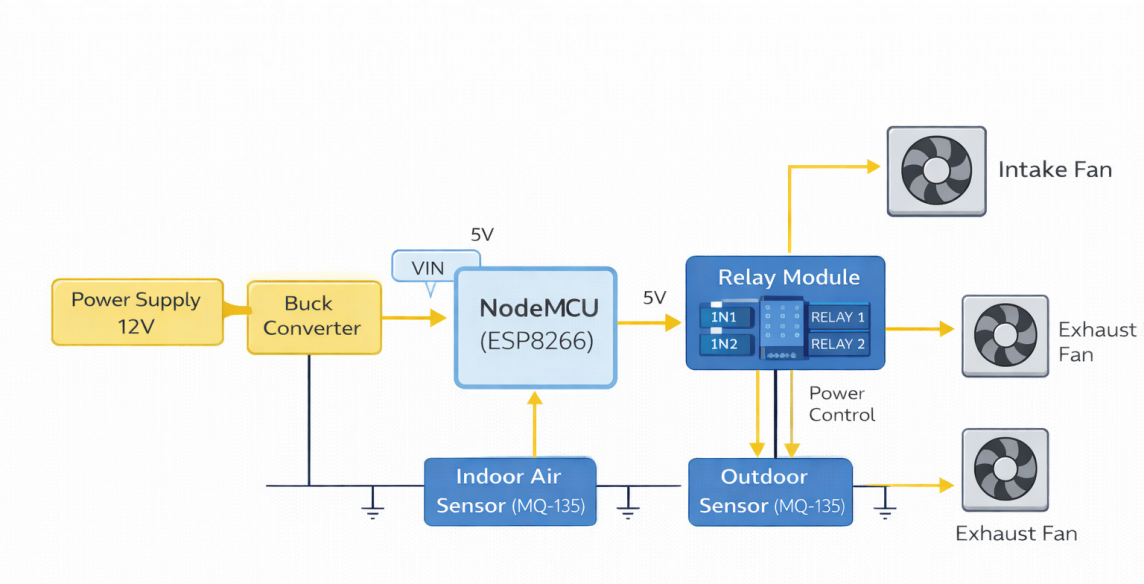


Figure 3.6: Power Distribution Diagram

Working of Power Distribution: The system is powered using a 12V DC power supply, which serves as the main source for all components. This voltage is stepped down to 5V using a buck converter to ensure safe operation of low-voltage components. The regulated 5V output is supplied to the NodeMCU through the VIN pin, as well as to the MQ-135 sensors and relay modules. The relay modules use this power to control the switching of ventilation fans, which may operate on higher voltage. A common ground is maintained across all components to ensure proper circuit operation and stable signal reference. This regulated power distribution ensures efficient and safe functioning of the entire system.

Sensor Interface Diagram

This diagram illustrates the connection between the MQ-135 gas sensors and the NodeMCU microcontroller used in the air quality monitoring system.

- Two MQ-135 sensors used for indoor and outdoor air quality detection
- Digital output pins of the sensors connected to NodeMCU GPIO pins
- Power connections provided from the regulated supply
- Common ground reference shared between sensors and the microcontroller

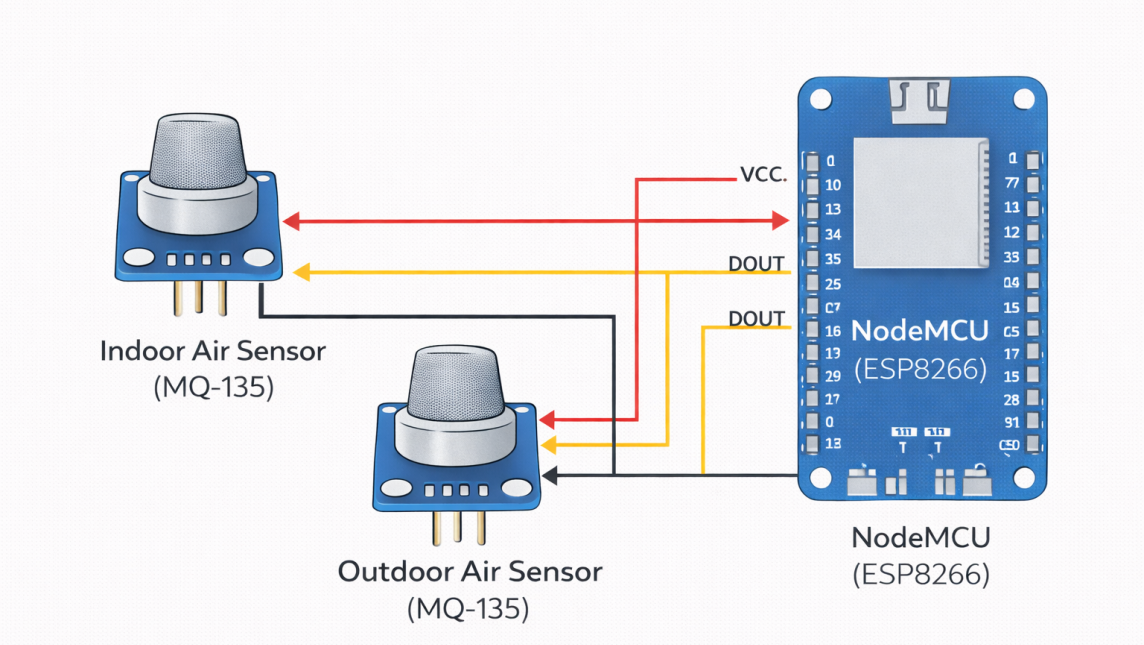


Figure 3.7: MQ-135 Sensor Connection Diagram

Working of Sensor Interface: The MQ-135 sensors continuously monitor air quality by detecting gases such as CO_2, NH_3, smoke, and other pollutants. In this system, two sensors are used to measure indoor and outdoor air conditions. The sensors generate output signals based on gas concentration, which are transmitted to the NodeMCU through GPIO pins. The NodeMCU reads these signals and processes them to determine the air quality levels. Based on this data, further actions such as ventilation control are performed. A regulated power supply ensures stable operation of the sensors, while a common ground provides a consistent reference for accurate signal transmission.

CHAPTER 4
IMPLEMENTATION

4.1 Hardware Assembly

The hardware assembly process involved integrating sensing, control, and ventilation components into a compact demonstration model. The system was constructed using a house-shaped enclosure that represents an indoor environment, allowing controlled air circulation and monitoring of indoor and outdoor air quality conditions. The main components integrated into the system include the NodeMCU microcontroller, MQ-135 gas sensors, relay modules, ventilation fans, and the power supply circuitry.

4.1.1 Mechanical Integration

A house model was constructed to simulate a real indoor environment. The structure was designed to allow controlled airflow and proper placement of sensors and ventilation components.

The model was used to mount the following components:

- NodeMCU microcontroller board
- Two MQ-135 gas sensors
- Relay modules
- Ventilation fans
- Buck converter
- 12V DC power supply connections

The ventilation fans were installed on the walls of the model to enable air circulation between indoor and outdoor spaces. One fan was positioned to exhaust polluted air, while the second fan allows fresh air to enter the indoor space.

The NodeMCU and other electronic components were placed outside the model structure to ensure safe wiring and easier maintenance.

4.1.2 Sensor Placement

Two MQ-135 gas sensors were used to detect air quality conditions in both indoor and outdoor environments. The sensors were positioned in the following configuration:

- **Indoor Sensor:** Placed inside the house model to detect indoor air quality conditions.
- **Outdoor Sensor:** Placed outside the house model to monitor external air conditions.

Proper positioning of the sensors was necessary to ensure reliable detection of air pollution and correct ventilation decisions.

4.1.3 Electrical Integration

The electrical connections were arranged carefully to ensure stable operation and reduce wiring complexity. The following integration steps were performed:

- MQ-135 sensor digital output pins connected to NodeMCU GPIO pins.
- Relay module input pins connected to NodeMCU control pins.
- Relay outputs connected to ventilation fans.
- Buck converter output connected to NodeMCU VIN pin.
- Sensors and relay modules powered through regulated voltage.
- A common ground connection established across all components.

The ventilation fans receive power from the 12V supply through relay modules, allowing the microcontroller to control airflow automatically.

4.1.4 Safety Considerations

The following precautions were taken during system implementation to ensure safe operation and prevent damage to electronic components:

- Buck converter output voltage was verified before connecting it to the NodeMCU.
- Proper insulation was maintained on all power supply connections.
- All ground connections were checked for continuity to avoid unstable operation.
- The system was powered **OFF** while making any wiring changes.

These safety measures ensured reliable system operation and protected the electronic components from potential electrical faults.

4.2 Software Implementation

The system software was developed using the Arduino IDE. The program was structured using modular functions to ensure readability and easier debugging.

4.2.1 Sensor Monitoring Logic

The NodeMCU continuously monitors digital outputs from the MQ-135 sensors to determine air quality conditions. The sensor readings indicate whether the air quality is safe or polluted.

Example logic:

```
if(indoorSensor == HIGH){
   // Indoor air safe
}
else{
   // Indoor air polluted
}
```

The microcontroller reads sensor signals repeatedly within the main control loop.

4.2.2 Ventilation Control Logic

Based on sensor readings, the system activates relay modules that control the ventilation fans.

The relay control functions include:

- Turn ON exhaust fan
- Turn ON intake fan
- Turn OFF ventilation system

These functions allow automated airflow control within the indoor environment.

4.2.3 System Operation

The main program loop performs the following sequence:

- Read indoor sensor status
- Read outdoor sensor status
- Evaluate air quality conditions
- Apply ventilation decision logic
- Activate or deactivate fans through relays
- Repeat continuously

The system operates in a reactive manner, meaning that ventilation decisions are made based on current sensor readings.

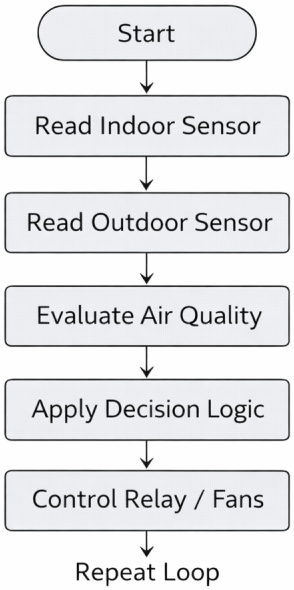


Figure 4.1: Software Execution Flow

4.3 System Design Decisions

During development, different sensing and control approaches were evaluated. However, certain practical constraints influenced the final system design.

4.3.1 Use of Digital Sensor Output

The MQ-135 sensor supports both analog and digital outputs. While analog output can provide gradual air quality measurements, the digital output mode was selected for this project.

Reasons include:

- Simplified signal processing
- Reduced software complexity
- Faster decision making

The digital output provides a simple indication of whether air quality is safe or polluted.

4.3.2 Ventilation-Based Air Quality Control

Instead of implementing complex air purification technologies, the system improves air quality through controlled ventilation.

This approach was selected because:

- It is cost effective
- It requires simple hardware components
- It demonstrates practical air circulation improvement

4.3.3 Final System Strategy

The final system uses a decision-based ventilation strategy where the NodeMCU continuously monitors air quality and automatically controls ventilation fans.

The system operates based on the following conditions:

- If indoor air is good → fans remain OFF
- If indoor air is polluted → ventilation fans activate
- If outdoor air is also polluted → filtered air circulation is used

4.3.4 Engineering Justification

The final implementation prioritizes simplicity, reliability, and cost efficiency. By combining sensor monitoring with automated ventilation control, the system provides a practical embedded solution for improving indoor air quality.

CHAPTER 5
TESTING

This chapter describes the procedures followed to validate individual components, subsystem integration, and overall system performance of the Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System.

5.1 Component Testing

Component testing was conducted to ensure that each hardware module functioned correctly before system-level integration.

5.1.1 Relay Module Testing

The relay modules were tested independently to verify correct switching behavior.

Procedure:

- The NodeMCU was programmed with a basic relay control test program.
- GPIO pins were toggled to send HIGH and LOW control signals to the relay module.
- The relay switching indicator LEDs were observed.
- The connected ventilation fan was monitored to verify correct ON and OFF operation.

Verification Criteria:

- Relay activates correctly when control signal is applied.
- Relay deactivates correctly when signal is removed.
- No abnormal heating or unstable switching occurs.
- Fan operation responds correctly to relay switching.

The test confirmed reliable relay switching and proper interfacing with the NodeMCU.

5.1.2 MQ-135 Sensor Testing

Each MQ-135 gas sensor was tested individually to verify correct air quality detection.

Procedure:

- The sensor was connected to the NodeMCU and powered through the regulated supply.
- Sensor digital output was monitored through the Serial Monitor.
- Different air conditions were created by exposing the sensor to smoke or gas sources.
- The sensor output response was observed.

Verification Criteria:

- Sensor output changes when exposed to polluted air.
- Sensor output remains stable in normal air conditions.
- No random signal fluctuations occur during testing.

The results confirmed that the MQ-135 sensors reliably detect changes in air quality.

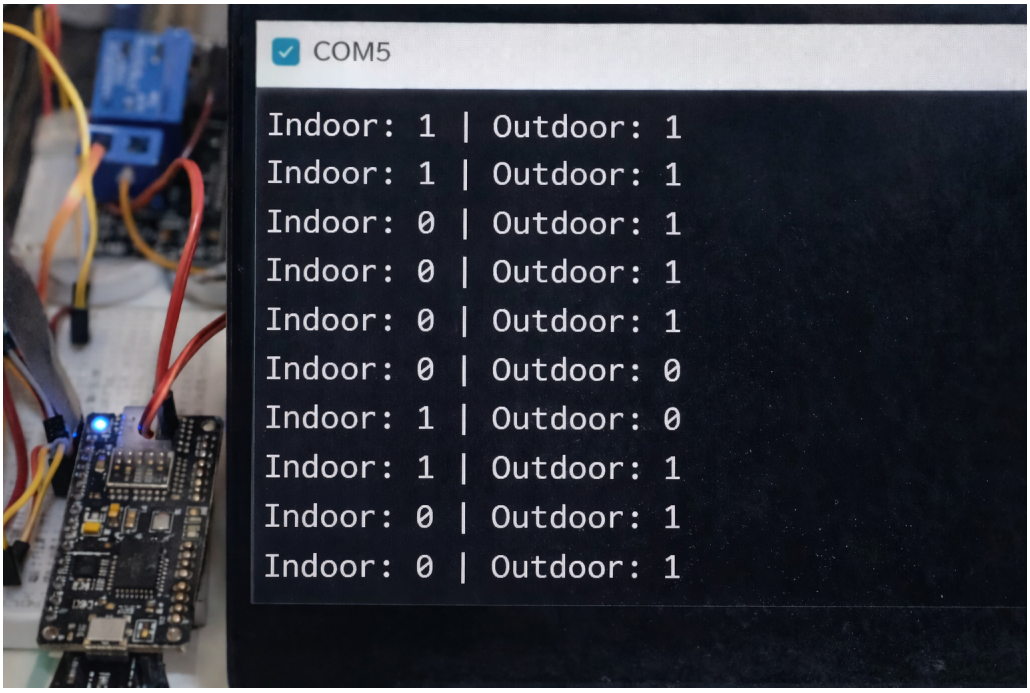


Figure 5.1: Serial Monitor Output During Sensor Testing

5.2 Integration Testing

After validating individual components, integration testing was performed to verify the coordination between sensors, controller, and ventilation system.

5.2.1 Sensor and Relay Integration

The integration between MQ-135 sensors and relay modules was tested to confirm correct automated control of ventilation fans.

Procedure:

- Indoor sensor readings were monitored continuously.
- Artificial smoke was introduced near the indoor sensor.
- The NodeMCU decision logic was observed.
- Relay activation and fan operation were verified.

The system successfully activated ventilation fans when polluted air was detected.

5.2.2 Common Ground Verification

To ensure signal stability, ground continuity tests were performed between all system components.

Checks performed:

- Continuity between NodeMCU ground and sensor ground.
- Relay module ground verification.
- Power supply ground verification.

Maintaining a common ground prevented unstable sensor readings and ensured reliable relay operation.

5.2.3 Power Stability Test

The system was powered using the 12V supply while all components were active.

The following conditions were tested:

- Sensors continuously monitoring air quality
- Relay modules switching ventilation fans
- Serial monitoring active for debugging

No unexpected resets or voltage instability were observed, confirming stable voltage regulation through the buck converter.

5.3 System Testing

System testing was performed to evaluate the complete operation of the air quality monitoring and ventilation control system.

5.3.1 Air Quality Detection Testing

The system was tested under different environmental conditions.

Test scenarios included:

- Normal indoor air conditions
- Indoor air polluted using smoke sources
- Outdoor air monitoring comparison

Expected system behavior:

- When indoor air is safe → ventilation fans remain OFF.
- When indoor air becomes polluted → ventilation fans activate.
- When both indoor and outdoor air are polluted → filtered air circulation occurs.

The system successfully detected air quality changes and responded accordingly.

5.3.2 Continuous Operation Testing

The system was operated continuously to evaluate stability and reliability.

Observations included:

- Stable sensor readings during extended runtime.
- Reliable relay switching without delays.
- Consistent ventilation fan operation.

The system maintained stable operation during prolonged testing periods.

5.3.3 Limitations Observed During Testing

During testing, the following limitations were observed:

- MQ-135 sensors provide trend detection rather than precise gas concentration measurements.
- Sensor response time may vary depending on environmental conditions.
- Digital sensor output only indicates polluted or safe air conditions without gradual measurement levels.

Despite these limitations, the system successfully demonstrated automatic monitoring and ventilation control.

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CHAPTER 6
CONCLUSION

The Embedded Indoor and Outdoor Air Quality Monitoring Ventilation System was successfully designed, assembled, programmed, and tested using a NodeMCU (ESP8266) based control system. The system integrates MQ-135 gas sensors, relay modules, ventilation fans, and a regulated power supply to monitor and control air quality conditions in a simulated indoor environment.

Two MQ-135 sensors were used to monitor indoor and outdoor air conditions. The NodeMCU processes the digital outputs from these sensors and determines whether ventilation is required. Based on the air quality conditions, relay modules control ventilation fans that regulate air circulation within the system. The system demonstrates automated ventilation control by activating fans when polluted air is detected and maintaining normal operation when air quality remains safe.

During the implementation and testing phases, several important observations were made. The MQ-135 sensors were effective in detecting changes in air quality when exposed to smoke or gas sources, making them suitable for low-cost embedded monitoring systems despite their limitation in providing precise gas concentration values. The use of a buck converter was essential for stepping down the 12V supply to a stable 5V level required by the NodeMCU and sensors, ensuring reliable operation and preventing damage to sensitive components.

Relay modules played a crucial role by enabling safe control of high-power ventilation fans through the low-voltage microcontroller, while maintaining electrical isolation between control and power circuits. It was also observed that proper placement of sensors and fans significantly influenced system performance, as correct positioning ensured accurate detection of air conditions and effective air circulation within the model.

Additionally, maintaining a common ground across all components was necessary for stable sensor readings and reliable relay operation, preventing signal fluctuations and improving overall system stability. The system maintained consistent performance during continuous operation, demonstrating reliability and effective coordination between sensing, control, and actuation components.

Overall, the project successfully demonstrates the integration of embedded systems, sensor monitoring, power regulation, and automated control mechanisms into a practical and cost-effective solution for improving indoor air quality through automated ventilation.

CHAPTER 7
FUTURE SCOPE

Although the current system successfully demonstrates automated air quality monitoring and ventilation control, several improvements can enhance system capability and performance.

Advanced Air Quality Sensors:

Future versions of the system could integrate more precise environmental sensors such as:

- NDIR CO_2 sensors
- Particulate matter sensors (PM2.5 / PM10)
- Multi-gas detection modules

These sensors would provide more accurate environmental monitoring.

IoT-Based Remote Monitoring:

The system can be enhanced by integrating cloud connectivity to allow remote monitoring of air quality through web or mobile applications. This would enable real-time data visualization and alerts.

Data Logging and Analysis:

Future systems could store air quality data over time to analyze environmental trends and improve ventilation control strategies.

Advanced Air Purification Methods:

Incorporating additional air purification technologies such as HEPA filters or air purifiers could further improve indoor air quality beyond basic ventilation.

Smart Building Integration:

The system could be integrated with smart building automation platforms to allow coordinated control of ventilation, air conditioning, and environmental monitoring systems.
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Appendices

The following code is used to monitor indoor and outdoor air quality using MQ-135 sensors and automatically control ventilation fans based on detected gas levels.

Source Code

```
#define INDOOR_SENSOR D5
#define OUTDOOR_SENSOR D6

#define INTAKE_FAN D1
#define EXHAUST_FAN D2

void setup() {
  Serial.begin(9600);

  pinMode(INDOOR_SENSOR, INPUT);
  pinMode(OUTDOOR_SENSOR, INPUT);

  pinMode(INTAKE_FAN, OUTPUT);
  pinMode(EXHAUST_FAN, OUTPUT);

  // Start OFF (Active LOW relay)
  digitalWrite(INTAKE_FAN, HIGH);
  digitalWrite(EXHAUST_FAN, HIGH);
}

void loop() {

  int indoor = digitalRead(INDOOR_SENSOR);
  int outdoor = digitalRead(OUTDOOR_SENSOR);

  Serial.print("Indoor: ");
  Serial.print(indoor);
  Serial.print(" | Outdoor: ");
  Serial.println(outdoor);

  // Gas detected (LOW = 0)
  if (indoor == 0) {
    digitalWrite(INTAKE_FAN, LOW);    // ON
    digitalWrite(EXHAUST_FAN, LOW);   // ON
  }

  // Indoor good but outdoor bad
  else if (outdoor == 0) {
    digitalWrite(INTAKE_FAN, HIGH);   // OFF
    digitalWrite(EXHAUST_FAN, HIGH);  // OFF
  }

  // Both good
```

```
  else {
    digitalWrite(INTAKE_FAN, HIGH);   // OFF
    digitalWrite(EXHAUST_FAN, HIGH);  // OFF
  }

  delay(500);
}
```